

Proof of Arnold Conjecture

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23.03.2026

1 Review

Firstly, let us review what the Arnold conjecture says and what we have done so far. Throughout, M is closed (compact without boundary), connected, and of dimension $2n$. We use

$$\iota_{X_{H_t}}\omega = dH_t, \quad g_J(v, w) = \omega(v, Jw), \quad \nabla_{g_J}H = -JX_H.$$

Thus J is ω -compatible and g_J is a Riemannian metric. The chosen sign of X_H also fixes the signs in the Floer equation below.

Theorem 1.1 (Arnold conjecture). *Let (M, ω) be a closed symplectic manifold and let $H : S^1 \times M \rightarrow \mathbb{R}$ be smooth, where $S^1 = \mathbb{R}/\mathbb{Z}$ and $H_t = H_{t+1}$. If every contractible 1-periodic solution of $\dot{x} = X_{H_t}(x)$ is nondegenerate, then*

$$\#\mathcal{P}_0(H) \geq \sum_{i=0}^{2n} \dim_{\mathbb{Q}} H_i(M; \mathbb{Q}).$$

Here $\mathcal{P}_0(H)$ denotes the contractible periodic solutions; nondegenerate means that 1 is not an eigenvalue of the linearized return map $d\phi_H^1(x(0))$. In particular the same lower bound holds when all periodic solutions are counted.

This is the nondegenerate Hamiltonian form of the conjecture. Its full generality requires analytical machinery beyond this seminar report. The direct Floer argument developed here is rigorous under the additional assumption $\omega|_{\pi_2(M)} = c_1(TM)|_{\pi_2(M)} = 0$, called *symplectic asphericity*. At the end we explain precisely what has to be supplied for monotone manifolds and cite the corresponding Floer–quantum homology result. This distinction is needed because compactness in the presence of holomorphic spheres cannot simply be inferred from an energy bound [1, 2]. The general theorem is established using additional virtual-cycle methods [5, 6].

The idea is similar to the Morse inequalities. We construct a chain complex whose generators are periodic solutions, compare its homology with the ordinary homology of M , and use the elementary inequality $\dim C_k \geq \dim H_k(C_*)$. For an aspherical manifold the contractible Floer complex has one generator per orbit, graded by its Conley–Zehnder index; its finite total dimension therefore bounds the total Betti number. Over $\mathbb{Q}$ this presupposes coherent orientations, which we include among the standard analytical ingredients of Floer theory. Alternatively the entire comparison can be made over $\mathbb{Z}_2$, giving the corresponding $\mathbb{Z}_2$ bound [1, 2].

By now, we have constructed Floer homology for a fixed regular pair of Hamiltonian H and compatible almost complex structure J on M . Its chain groups are generated by the contractible 1-periodic solutions, with the preceding conventions on cappings and coefficients. What remains is to show that its homology agrees with Morse homology for a convenient small Hamiltonian, and then

that it is independent of the chosen regular pair. We first examine the ordinary differential operators behind this comparison; afterward we construct continuation maps.

For clarity, an orbit is called contractible when its loop in M bounds a smooth disk $w : D^2 \rightarrow M$. In a nonaspherical manifold, two such disks can differ by a sphere class $A \in \pi_2(M)$; the symplectic area changes by $\int_A \omega$ and the grading changes by twice $\langle c_1(TM), A \rangle$. Our asphericity assumption removes both ambiguities, so the action and integer grading depend only on the loop. Moreover, a nondegenerate 1-periodic orbit is an isolated fixed point of the time-one Hamiltonian map. Since the fixed-point set is a closed subset of compact M , a set of isolated fixed points is finite in this nondegenerate situation: an infinite set would have an accumulation point that is also fixed and hence not isolated. This finiteness makes the elementary dimension argument at the end literally a finite-dimensional one.

The Floer differential counts solutions of an elliptic equation on $\mathbb{R} \times S^1$. To obtain a chain complex one needs more than a formal count: index-one solutions modulo translation must form finite sets, and compactified index-two families must have exactly the broken pairs that prove $\partial^2 = 0$. Transversality makes their expected dimensions actual dimensions; energy estimates and compactness control their limits; gluing describes neighborhoods of broken limits. These facts were part of the previous construction of Floer homology, and we will use the same ingredients when the equation depends on the continuation parameter s . Stating them here also indicates which claims in the later proofs rely on external Floer analysis.

2 From Floer to Morse

Firstly, we show that a sufficiently small autonomous Morse function has only constant 1-periodic Hamiltonian orbits. The word “small” will refer to a fixed Riemannian C^2 norm; compactness of M allows all local estimates below to be made uniform.

Proposition 2.1. *Let $H : M \rightarrow \mathbb{R}$ be a Morse function. For $\varepsilon > 0$ sufficiently small, every 1-periodic orbit of $X_{\varepsilon H}$ is constant, and hence $\mathcal{P}(\varepsilon H) = \text{Crit}(H)$.*

Proof. Choose a finite collection of Darboux charts whose smaller concentric charts still cover M . Since $\|X_{\varepsilon H}\|_{C^0} = O(\varepsilon)$, the length of any such orbit is at most $C\varepsilon$. For small ε , its image lies in one of the larger charts. This justifies using a single Euclidean coordinate system along the entire orbit; the chart depends on the orbit, but the constants can be chosen uniformly from our finite cover.

Write the coordinate expression of the orbit as a periodic map $x : S^1 \rightarrow \mathbb{R}^{2n}$, and suppose $\|DX_{\varepsilon H}\| < 2\pi$ throughout the relevant charts. Its derivative has zero mean. With the Fourier expansion

$$x(t) = \sum_{m \in \mathbb{Z}} c_m e^{2\pi i m t}, \quad x'(t) = \sum_{m \neq 0} 2\pi i m c_m e^{2\pi i m t}, \quad x''(t) = \sum_{m \neq 0} (2\pi i m)^2 c_m e^{2\pi i m t},$$

Parseval's identity gives the periodic Poincaré inequality

$$\|x''\|_{L^2}^2 = \sum_{m \neq 0} (2\pi m)^4 |c_m|^2 \geq (2\pi)^2 \sum_{m \neq 0} (2\pi m)^2 |c_m|^2 = (2\pi)^2 \|x'\|_{L^2}^2.$$

On the other hand, differentiating $x' = X_{\varepsilon H}(x)$ gives

$$x'' = DX_{\varepsilon H}(x)x', \quad \|x''\|_{L^2} < 2\pi \|x'\|_{L^2}$$

unless $x' = 0$. These two inequalities contradict each other if x' is nonzero. Therefore the orbit is constant, and $X_{\varepsilon H}(x) = 0$ is equivalent to $dH_x = 0$. $\square$

Thus the *generators* of the Floer complex can be identified with the critical points of H . This does not yet show that the Floer and Morse *differentials* coincide: we also have to identify their gradings, orientations, and trajectories. In the aspherical case cappings do not change the action or the grading. For a nonaspherical manifold the generators must instead be capped or recorded over a Novikov ring, and the bare equality $CF_* = CM_*$ is generally false.

Proposition 2.2. *Use the Conley–Zehnder convention in which the path $\exp(tJ_0S)$, for a sufficiently small nonsingular symmetric matrix S , has index $\mu^-(S) - n$. Then, for every $x \in \text{Crit}(H)$ and small $\varepsilon > 0$,*

$$\mu_{\text{CZ}}(x) = \text{ind}_H(x) - n.$$

Here $\text{ind}_H(x)$ is the number of negative eigenvalues of the Hessian of H at x .

Proof. A constant orbit has the constant disk as a capping. Choose symplectic coordinates at x with ω standard. Since $X_H = J_0 \nabla H$ in these coordinates, the linearized Hamiltonian flow is

$$\Psi_x(t) = \exp(tJ_0\varepsilon \text{Hess}_x H), \quad 0 \leq t \leq 1.$$

The Hessian is nonsingular because H is Morse. Shrinking ε ensures that $\Psi_x(1)$ has no eigenvalue 1: near the identity the only possible crossing is at $t = 0$. With the indicated convention the crossing form there is the quadratic form of $\varepsilon \text{Hess}_x H$ and its half-signature gives

$$\mu_{\text{CZ}}(\Psi_x) = -\frac{1}{2} \text{sign}(\text{Hess}_x H) = \mu^-(\text{Hess}_x H) - n.$$

The equality in the last line uses $\dim M = 2n$. Changing the convention for X_H or the Conley–Zehnder index changes this displayed shift, so we keep both conventions fixed throughout [3, 2]. $\square$

Thus the Floer and Morse gradings agree after shifting by n . In particular, $CF_k(\varepsilon H, J)$ corresponds to $CM_{k+n}(H)$. To compare the differentials, choose a compatible J for which the negative g_J gradient flow of H is Morse–Smale. Our equations, with $X = \nabla_{g_J} H = -JX_H$, are

$$\text{Morse:} \quad \frac{du}{ds} + X(u) = 0, \tag{1}$$

$$\text{Floer:} \quad \partial_s u + J(u)(\partial_t u - X_H(u)) = 0. \tag{2}$$

For an autonomous H , a t -independent solution of (2) satisfies (1). We denote the section in (1) by $\bar{\partial}_X(u) = u' + X(u)$ and its linearization along a trajectory u by D_u .

In a trivialization of u^*TM adapted to g_J , the operator takes the form

$$D_u Y = Y' + A(s)Y : W^{1,2}(\mathbb{R}, \mathbb{R}^{2n}) \longrightarrow L^2(\mathbb{R}, \mathbb{R}^{2n}),$$

where $A(s)$ tends to symmetric invertible matrices $A_\pm$ as $s \rightarrow \pm\infty$. The limiting matrices represent the Riemannian Hessians at the critical endpoints $x^\pm$. A choice of connection or trivialization may change the matrix $A(s)$ in the interior, but not the Fredholm or index assertions.

Proposition 2.3. *If H is Morse and $u : \mathbb{R} \rightarrow M$ solves (1) with $u(s) \rightarrow x^\pm \in \text{Crit}(H)$ as $s \rightarrow \pm\infty$, then $D_u : W^{1,2}(\mathbb{R}, u^*TM) \rightarrow L^2(\mathbb{R}, u^*TM)$ is Fredholm.*

Proof. Recall that Fredholm means bounded with finite-dimensional kernel and cokernel and closed image. Boundedness follows immediately from $A \in L^\infty$:

$$\|D_u Y\|_{L^2} \leq \|Y'\|_{L^2} + \|A\|_{L^\infty} \|Y\|_{L^2} \leq C \|Y\|_{W^{1,2}}.$$

We first obtain a uniform estimate at each end. Set $L_{\pm} = d/ds + A_{\pm}$ on the whole line. Fourier transformation gives

$$\widehat{L_{\pm}Y}(\xi) = (i\xi + A_{\pm})\widehat{Y}(\xi).$$

If $\lambda_1, \dots, \lambda_{2n}$ are the real nonzero eigenvalues of $A_{\pm}$, then

$$|i\xi + \lambda_j|^2 = \xi^2 + \lambda_j^2 \geq c(1 + \xi^2)$$

for some $c > 0$ independent of ξ and j . In particular $i\xi + A_{\pm}$ is invertible for all real ξ , and Plancherel's theorem yields

$$\|Y\|_{W^{1,2}}^2 = \int_{\mathbb{R}} (1 + \xi^2) |\widehat{Y}(\xi)|^2 d\xi \leq C \int_{\mathbb{R}} |\widehat{L_{\pm}Y}(\xi)|^2 d\xi = C \|L_{\pm}Y\|_{L^2}^2. \quad (3)$$

This also shows explicitly why nondegeneracy of the endpoints is needed.

Since $A(s) \rightarrow A_{\pm}$, choose T so that $\|A(s) - A_{\pm}\|$ is as small as desired on the corresponding ends. Apply (3) to cutoff sections supported in $s > T$ and $s < -T$. Their cutoff derivatives are supported in a fixed compact interval. Absorbing the small end terms and using the ordinary local $W^{1,2}$ estimate in the middle gives

$$\|Y\|_{W^{1,2}(\mathbb{R})} \leq C(\|D_u Y\|_{L^2(\mathbb{R})} + \|Y\|_{L^2([-T-1, T+1])}). \quad (4)$$

We may see more concretely why the compact term in (4) is local. Let β_+ vanish before T and equal 1 after $T+1$. Then

$$L_+(\beta_+ Y) = \beta_+ D_u Y + \beta'_+ Y + \beta_+(A_+ - A(s))Y.$$

Applying (3) to $\beta_+ Y$, the last term is absorbed into the left side when $\sup_{s>T} \|A(s) - A_+\|$ is sufficiently small. The derivative term $\beta'_+ Y$ lives only in $[T, T+1]$. The analogous identity with a cutoff β_- handles the negative end. A third cutoff covers $[-T-1, T+1]$, where

$$\|Y\|_{W^{1,2}([-T, T])} \leq C(\|D_u Y\|_{L^2([-T-1, T+1])} + \|Y\|_{L^2([-T-1, T+1])}).$$

Adding these three estimates gives (4); the constants do not depend on Y . Thus the Fourier calculation on a constant-coefficient whole-line operator really does control the variable-coefficient operator at infinity.

The same argument applies to the formal L^2 adjoint

$$D_u^* = -\frac{d}{ds} + A(s)^T : W^{1,2}(\mathbb{R}, \mathbb{R}^{2n}) \longrightarrow L^2(\mathbb{R}, \mathbb{R}^{2n}); \quad (5)$$

its limiting matrices are $A_{\pm}^T$ and remain invertible.

For $Y \in \ker D_u$, estimate (4) becomes

$$\|Y\|_{W^{1,2}} \leq C\|Y\|_{L^2([-T-1, T+1])}. \quad (6)$$

Restriction $W^{1,2}(\mathbb{R}) \rightarrow L^2([-T-1, T+1])$ is compact by the Rellich compact embedding on a bounded interval. In one dimension this compactness can also be seen directly. For a uniformly $W^{1,2}$ -bounded sequence Y_j on $[-T-1, T+1]$, the absolutely continuous representatives satisfy

$$|Y_j(s) - Y_j(t)| \leq \int_t^s |Y_j'(r)| dr \leq |s - t|^{1/2} \|Y_j'\|_{L^2([-T-1, T+1])}.$$

The sequence is uniformly bounded there as well, by the one-dimensional Sobolev embedding. Arzelà–Ascoli then gives a uniformly convergent subsequence and hence an L^2 -convergent one. This elementary route explains the compact restriction used below without requiring compactness of the embedding on the entire noncompact real line. If a sequence in the unit ball of $\ker D_u$ has restrictions converging in L^2 , applying (6) to differences makes it converge in $W^{1,2}(\mathbb{R})$. Thus that unit ball is compact and $\ker D_u$ is finite-dimensional. The identical argument for D_u^* shows $\ker D_u^*$ is finite-dimensional.

For completeness, closed range requires an extra argument; (4) alone does not bound a general sequence Y_j . It implies the stronger estimate

$$\|Y\|_{W^{1,2}} \leq C' \|D_u Y\|_{L^2}, \quad Y \perp \ker D_u.$$

Indeed, otherwise there are unit vectors $Y_j \perp \ker D_u$ with $D_u Y_j \rightarrow 0$. A subsequence converges weakly in $W^{1,2}$ and strongly on the compact interval by Rellich. Estimate (4) applied to differences makes the subsequence converge strongly in $W^{1,2}$, to a unit vector in $\ker D_u$ perpendicular to that kernel, a contradiction. Now if $D_u Y_j \rightarrow Z$, subtract the orthogonal projections of Y_j onto $\ker D_u$; the stronger estimate shows the resulting sequence is Cauchy, so Z belongs to the image. Finally $(\operatorname{im} D_u)^\perp = \ker D_u^*$, and the now closed image has finite-dimensional cokernel. This proves all four Fredholm requirements [2]. $\square$

After this, we compute the Fredholm index. We retain the stable and unstable subspace interpretation because it will also explain transversality.

Proposition 2.4. *For a trajectory from x^- to x^+ ,*

$$\operatorname{index}(D_u) = \operatorname{ind}_H(x^-) - \operatorname{ind}_H(x^+).$$

Proof. For a fixed σ , solve $Z' + A(s)Z = 0$ with $Z(\sigma) = v$. This defines an evolution map $\Phi(\sigma, s) : T_{u(\sigma)}M \rightarrow T_{u(s)}M$. Define

$$E^u(\sigma) = \{v : \Phi(\sigma, s)v \rightarrow 0 \text{ as } s \rightarrow -\infty\}, \quad E^s(\sigma) = \{v : \Phi(\sigma, s)v \rightarrow 0 \text{ as } s \rightarrow +\infty\}.$$

The stable-manifold theorem for the negative gradient flow identifies these spaces with $T_{u(\sigma)}W^u(x^-)$ and $T_{u(\sigma)}W^s(x^+)$, respectively. Their dimensions are

$$\dim E^u = \operatorname{ind}_H(x^-), \quad \dim E^s = 2n - \operatorname{ind}_H(x^+).$$

An L^2 solution of $D_u Y = 0$ decays at both ends, so evaluation at σ identifies $\ker D_u$ with $E^u \cap E^s$.

The adjoint equation is $-\eta' + A(s)^T \eta = 0$. For its solutions, $\langle \eta(s), Y(s) \rangle$ is constant whenever $D_u Y = 0$. Decay at $+\infty$ makes an adjoint L^2 solution orthogonal to E^s , and decay at $-\infty$ makes it orthogonal to E^u . Conversely, the exponential dichotomies of the limiting nonsingular matrices show that a vector orthogonal to $E^u + E^s$ extends to an L^2 adjoint solution. Hence

$$\ker D_u^* \cong (E^u + E^s)^\perp, \quad \dim \operatorname{coker} D_u = 2n - \dim(E^u + E^s).$$

It is the orthogonal complement of the *sum*, rather than of the intersection, that occurs here. Using $\dim(E^u \cap E^s) + \dim(E^u + E^s) = \dim E^u + \dim E^s$, we obtain

$$\begin{aligned} \operatorname{index}(D_u) &= \dim(E^u \cap E^s) - [2n - \dim(E^u + E^s)] \\ &= \dim E^u + \dim E^s - 2n = \operatorname{ind}_H(x^-) - \operatorname{ind}_H(x^+). \end{aligned}$$

The constant trajectory has the same formula, either directly from the limiting operator or by a separate local calculation. $\square$

Proposition 2.5. *The negative gradient flow of (H, g_J) is Morse–Smale if and only if D_u is surjective for every trajectory u between critical points (including the relevant constant trajectories).*

Proof. By Proposition 2.3, D_u has closed image, so surjectivity is equivalent to $\ker D_u^* = 0$. The identification just obtained says this is equivalent to

$$E^u(\sigma) + E^s(\sigma) = T_{u(\sigma)}M,$$

or $T_{u(\sigma)}W^u(x^-) + T_{u(\sigma)}W^s(x^+) = T_{u(\sigma)}M$. This is precisely transverse intersection of the stable and unstable manifolds along u , the Morse–Smale condition. $\square$

Now we consider the Floer part. We need regularity of the linearized Floer operator along the t -independent Morse trajectories. The argument comparing kernels is clearest in a compatible J adapted near the relevant trajectories; in arbitrary moving coordinates the derivative of J and the connection terms must be retained.

Proposition 2.6. *For a closed M and a Morse–Smale pair (H, g_J) , the linearized Floer operator along every t -independent solution for εH is surjective when $\varepsilon > 0$ is sufficiently small. Its kernel consists of the t -independent kernel of the Morse linearization, after the rescaling $s \mapsto \varepsilon s$.*

Sketch of proof. Along a gradient line $u(s)$, split a section $\xi(s, t)$ into its t -average $\bar{\xi}(s)$ and its zero-mean part $\xi_0(s, t)$. For a constant compatible complex structure in a local symplectic trivialization, the model operator is

$$D_u^F \xi = \partial_s \xi + J_0 \partial_t \xi + \varepsilon A(s) \xi, \quad D_u^M \bar{\xi} = \partial_s \bar{\xi} + \varepsilon A(s) \bar{\xi}.$$

Averaging $D_u^F \xi = 0$ over t yields $D_u^M \bar{\xi} = 0$. Conversely a solution of the latter gives a t -independent solution of the former. To rule out a zero-mean solution, the Fourier Poincaré inequality gives

$$\|\xi_0\|_{L^2(\mathbb{R} \times S^1)} \leq (2\pi)^{-1} \|\partial_t \xi_0\|_{L^2(\mathbb{R} \times S^1)}.$$

Integration by parts, periodicity in t , and decay at the ends remove the cross term in

$$\|\partial_s \xi_0 + J_0 \partial_t \xi_0\|_{L^2}^2 = \|\partial_s \xi_0\|_{L^2}^2 + \|\partial_t \xi_0\|_{L^2}^2.$$

Thus $D_u^F \xi_0 = 0$ implies

$$\|\partial_t \xi_0\|_{L^2} \leq \varepsilon \|A\|_{L^\infty} \|\xi_0\|_{L^2} \leq \frac{\varepsilon \|A\|_{L^\infty}}{2\pi} \|\partial_t \xi_0\|_{L^2},$$

forcing $\xi_0 = 0$ for small ε . The corresponding estimate for the formal adjoint removes its nonconstant Fourier modes. Since the Morse adjoint has zero kernel by Proposition 2.5, the Floer adjoint does too, giving surjectivity.

For a general compatible J along the trajectory, averaging in a fixed coordinate frame need not commute with the operator, and the cross term above is not literally zero. One uses a unitary connection and the spectral gap of the nonconstant modes; the extra J and connection terms are controlled by the $O(\varepsilon)$ variation of the rescaled gradient line, uniformly on the compact manifold. This perturbation and the exponential end estimates constitute the standard small-Hamiltonian comparison theorem [2, Sec. 2.7]. We use the conclusion of that theorem here, rather than claiming the constant-coefficient calculation by itself proves the general case. $\square$

In particular, Morse–Smale transversality gives the required Floer regularity *along these solutions*. We still need to know that trajectories counted in the differential do not depend on t .

Notice the two logically different questions here. Surjectivity of the operator at a known gradient cylinder tells us that the corresponding Floer moduli space is smooth near that cylinder. It does not rule out other cylinders elsewhere in the moduli space. The small-Hamiltonian correspondence uses compactness to bring every low-index cylinder close to a broken collection of gradient cylinders, and only then uses regularity and local uniqueness. Keeping these steps separate prevents the equality of kernels from being mistaken for a global classification of all solutions.

Proposition 2.7. *Assume (M, ω) is symplectically aspherical, H is Morse, and (H, g_J) is Morse–Smale. For sufficiently small $\varepsilon > 0$, every Floer trajectory of εH with Conley–Zehnder index difference 1 is t -independent. The same low-index comparison, with broken trajectories at its boundary, applies to index difference 2.*

Sketch of proof. The original rescaling idea must be used with care. If u solves the equation for H/k , then $v(s, t) = u(ks, kt)$ formally solves the equation for H , but $t \mapsto kt$ is a multiple cover of the circle. Its energy and end behavior are not controlled by compactness for a fixed pair of orbits in the way needed for a limit of the proposed kind. In particular, convergence to one Morse trajectory does not follow from that rescaling.

The small-Hamiltonian theorem instead starts with the energy identity and compactness on the *original* cylinder. In the aspherical case, for fixed endpoints $x^\pm$,

$$E(u) = \int_{\mathbb{R} \times S^1} |\partial_s u|^2 ds = \mathcal{A}_{\varepsilon H}(x^-) - \mathcal{A}_{\varepsilon H}(x^+) = \varepsilon[H(x^-) - H(x^+)],$$

where the action convention is $\mathcal{A}_H([x, w]) = -\int_{D^2} w^* \omega + \int_0^1 H_t(x(t)) dt$. Hence $E(u) = O(\varepsilon)$. The mean-value inequality and a blow-up argument exclude energy concentration; any nonconstant sphere bubble would have positive energy bounded away from zero for fixed tame geometry, while asphericity excludes such spheres entirely. Elliptic estimates, followed by a suitable rescaling of the s variable only, show that a sequence of these cylinders approaches a chain of negative gradient lines. Translation in s may be needed before each nonconstant level converges.

Near each regular gradient line the implicit function theorem and the uniform inverse estimate on the nonconstant t modes from Proposition 2.6 identify the nearby Floer trajectories with the t -independent gradient trajectories. The local uniqueness is essential: convergence to a gradient line alone would not force a nearby solution to be independent of t . For index difference 1, compactness and the index formula leave one unbroken line up to translation. At index difference 2, the compactification also contains pairs of index-one lines; the same local comparison and gluing describe their neighborhoods. This is the standard Floer–Morse correspondence for a sufficiently small autonomous Hamiltonian [1, 2]. The compactness, local uniqueness, and gluing statements used here are substantial analytic results, so this is a sketch rather than a complete proof.

The index bound matters in this compactness argument. An unbroken nonconstant negative gradient trajectory has a free translation parameter, so after dividing by translation its dimension is $\text{ind}_H(x^-) - \text{ind}_H(x^+) - 1$. A pair of nonconstant lines has one additional breaking parameter in the compactification, and it first appears as the boundary of a one-dimensional quotient when the index difference is 2. For index difference 1 a pair of positive-index lines cannot occur. This dimension count identifies the strata needed for the differential and for $\partial^2 = 0$; it is not a general exclusion of all time-dependent high-index trajectories. Likewise, a solution of index difference 0 between distinct critical points cannot be a nonconstant Morse line for a Morse–Smale pair. These restrictions are what permit the low-index identification used in the homology calculation. $\square$

We also need an appropriate compatible almost complex structure. It is useful to state the existence result accurately: it is a property of the metric g_J , and varying J does not produce every Riemannian metric.

Theorem 2.8. *Let (M, ω) be closed and $H : M \rightarrow \mathbb{R}$ Morse. In the space $\mathcal{J}(M, \omega)$ of smooth compatible almost complex structures, the set of J for which the negative g_J gradient flow of H is Morse–Smale is residual, hence dense.*

Sketch of proof. By Proposition 2.5, we seek surjectivity of D_u on every connecting trajectory. Complete the relevant spaces of maps $u : \mathbb{R} \rightarrow M$ with exponential convergence at the critical endpoints and of compatible structures J in suitable Sobolev or C^ℓ norms. Over the space of pairs (u, J) define the section

$$\mathcal{F}(u, J) = u' - J(u)X_H(u).$$

Its vertical differential, at a zero, is

$$D\mathcal{F}_{(u, J)}(\xi, Y) = D_u\xi - Y(u)X_H(u),$$

where Y is tangent to the space of compatible almost complex structures: $YJ + JY = 0$ and $\omega(\cdot, Y\cdot)$ is symmetric. The first summand is Fredholm by Proposition 2.3.

Suppose a nonzero $\eta \in \ker D_u^*$ annihilates also every variation $Y(u)X_H(u)$. Choose a point s_0 on a nonconstant trajectory where $\eta(s_0) \neq 0$. Such a point exists because the adjoint is an ODE and a nonzero solution cannot vanish at one point. Also $X_H(u(s_0)) \neq 0$: a nonconstant gradient line cannot reach a critical point at finite time. The line $u(\mathbb{R})$ is injective by strict decrease of H , so a variation Y supported near $u(s_0)$ can be localized to a short interval of the trajectory. The tangent variations of compatible J supply a nonzero pairing with $\eta(s_0)$ after a choice of local variation; in the few algebraic directions not detected by this particular variation, the adjoint equation and the flow direction give the required exclusion. This local surjectivity calculation, including that algebraic point, is a standard transversality lemma [2, 4]. We cannot infer it merely from $X_H \neq 0$, since Y is constrained by compatibility.

Consequently the universal section is transverse to zero. Its zero set is a Banach manifold; the projection to the space of compatible J is a Fredholm map. Sard–Smale implies that its regular values form a residual set. Taking the countable intersection over critical-point pairs and the required differentiability classes, then using elliptic regularity, gives smooth compatible structures with all stable and unstable intersections transverse. Constant trajectories are automatically regular because the Hessian at a critical point is nonsingular. This proves the stated residual conclusion subject to the standard Banach-completion and transversality lemma just cited. $\square$

Combining the generator identification, Proposition 2.2, the index-one trajectory comparison, and coherent orientations gives, for sufficiently small ε ,

$$CF_k(\varepsilon H, J; \mathbb{Q}) \cong CM_{k+n}(H, g_J; \mathbb{Q}), \quad \partial_F = \partial_M, \quad HF_k(\varepsilon H, J; \mathbb{Q}) \cong H_{k+n}(M; \mathbb{Q}).$$

The equality of differentials includes the agreement of signs after compatible choices of orientation lines. Over $\mathbb{Z}_2$ it requires no sign convention. For aspherical M , all periodic orbits of εH are constant by Proposition 2.1, so this is an identification of the entire contractible complex.

3 Invariance of Floer Homology

Now the remaining part of the Arnold argument is the invariance of Floer homology. We give the continuation construction and the dimensions of its moduli spaces. In this section first assume

symplectic asphericity; for arbitrary H_t choose regular time-dependent compatible J_t . The Floer differential counts oriented isolated cylinders modulo translation. We use the regularity, compactness up to breaking, and gluing theorems of Hamiltonian Floer theory, rather than claiming these follow from the Fredholm calculation alone [1, 2].

Theorem 3.1. *Let (M, ω) be closed and symplectically aspherical. If (H^-, J^-) and (H^+, J^+) are regular Floer data, there is a canonical isomorphism*

$$\Phi_{-+} : HF_*(H^-, J^-; \mathbb{Q}) \longrightarrow HF_*(H^+, J^+; \mathbb{Q}).$$

The maps are independent of the chosen continuation data on homology and respect composition.

Sketch of proof. Choose an asymptotically constant path χ from the minus data to the plus data. Count index-zero solutions of the s -dependent Floer equation to obtain $\Phi_\chi^\#$. The boundary of the compactified one-dimensional continuation moduli space yields $\partial^+ \Phi_\chi^\# = \Phi_\chi^\# \partial^-$. A generic homotopy of two paths gives a one-parameter moduli space; its index-minus-one points define a chain homotopy, so the induced maps agree. Gluing two continuation paths produces a map chain homotopic to the composition, and a constant path induces the identity. The continuation along a reversed path is therefore inverse on homology. We spell out each of these steps below, including the endpoint and sign conventions. $\square$

Definition 3.2. Let $(H^\pm, J^\pm)$ be regular. An asymptotically constant continuation datum $\chi = (H_t^s, J_t^s)$ is a smooth family equal to (H^-, J^-) for $s \leq -T$ and to (H^+, J^+) for $s \geq T$, for some $T > 0$. Write $\mathcal{P}(H^-, J^-; H^+, J^+)$ for its space of choices. No regularity of an arbitrary member of $\mathcal{P}$ is assumed.

The associated section is

$$\bar{\partial}_\chi(u) = \partial_s u + J_t^s(u)(\partial_t u - X_{H_t^s}(u)).$$

For $\gamma^\pm \in \mathcal{P}_0(H^\pm)$, define

Definition 3.3.

$$\mathcal{N}(\gamma^-, \gamma^+; \chi) = \left\{ u : \mathbb{R} \times S^1 \rightarrow M : \bar{\partial}_\chi(u) = 0, \quad \lim_{s \rightarrow \pm\infty} u(s, \cdot) = \gamma^\pm \right\}.$$

The limits are smooth and uniform in t . On a nonaspherical manifold one also specifies compatible endpoint cappings and the relative homotopy class of u .

Let $D_{u, \chi}$ be the linearization of this section on suitable weighted Sobolev spaces. In a trivialization along u , its principal part is $\partial_s + J_t^s \partial_t$; the remaining zeroth-order terms come from ∇J_t^s , $\nabla X_{H_t^s}$, and the chosen connection. The asymptotic operators at $s = \pm\infty$ are invertible because the orbits are nondegenerate. The index theorem for such Cauchy–Riemann type operators computes the Fredholm index from the spectral flow, equivalently the difference of endpoint Conley–Zehnder indices. Changing the continuation path in the middle does not change this index, provided the endpoints and the relevant capping class remain fixed. This is the two-dimensional analogue of the limiting-operator and stable-subspace discussion in Propositions 2.3 and 2.4.

Consequently $D_{u, \chi}$ is Fredholm, and in the aspherical setting its index is

$$\text{index}(D_{u, \chi}) = \mu_{\text{CZ}}(\gamma^-) - \mu_{\text{CZ}}(\gamma^+).$$

Unlike an autonomous Floer equation, a genuinely s -dependent continuation equation has no $\mathbb{R}$ translation symmetry: translating u changes the datum H^s, J^s . Its expected moduli-space dimension is therefore the Fredholm index itself, with no subtraction of one. Generic perturbations supported away from the ends make the section transverse to zero; this follows from the universal moduli-space Sard–Smale argument for Floer continuation data [2]. We call such a datum regular and denote its space by $\mathcal{P}_{\text{reg}}$.

Energy remains central, but the continuation equation has an additional term. With our action convention and compatible cappings,

$$E(u) := \int_{\mathbb{R} \times S^1} |\partial_s u|_{g_{J_t^s}}^2 ds dt = \mathcal{A}_{H^-}(\gamma^-) - \mathcal{A}_{H^+}(\gamma^+) + \int_{\mathbb{R} \times S^1} (\partial_s H_t^s)(u) ds dt. \quad (7)$$

Indeed the s derivative of $\mathcal{A}_{H^s}(u(s, \cdot))$ equals $-\|\partial_s u\|^2 + \int_{S^1} \partial_s H_t^s(u) dt$. Integrating gives (7). Since $\partial_s H^s$ has compact s support,

$$E(u) \leq \mathcal{A}_{H^-}(\gamma^-) - \mathcal{A}_{H^+}(\gamma^+) + \int_{\mathbb{R}} \max_{t,x} \partial_s H_t^s(x) ds.$$

There are finitely many nondegenerate contractible periodic orbits at the two ends, and in the aspherical setting their actions are unambiguous. Thus a single datum gives a uniform bound over all its finite-energy connecting cylinders. To derive the sign in (7) explicitly, the derivative of the action along a cylinder is

$$\frac{d}{ds} \mathcal{A}_{H^s}(u(s, \cdot)) = - \int_{S^1} \omega(\partial_s u, \partial_t u - X_{H_t^s}) dt + \int_{S^1} \partial_s H_t^s(u) dt.$$

The equation gives $\partial_t u - X_{H_t^s} = J_t^s \partial_s u$, so the first integral becomes $-\int_{S^1} |\partial_s u|^2 dt$. Integrating between $-S$ and S and passing to the asymptotic limits yields the stated formula. If H^s were independent of s , the last term would vanish and we would recover the ordinary action-difference identity. The term is indispensable for continuation: it is the price of moving the action functional itself.

The bound by itself does *not* say that the whole moduli space is compact: sequences may break at either end. Asphericity excludes nonconstant sphere bubbles, and the Floer–Gromov compactness theorem gives compactness *up to breaking*; in expected dimensions zero and one the broken configurations have the forms used below. In particular the regular zero-dimensional space $\mathcal{N}_0(\gamma^-, \gamma^+; \chi)$ is finite. The original assertion that every finite-energy continuation moduli space is compact without adding broken trajectories was too strong.

Choose coherent orientations, and denote by $\epsilon^\chi(u) \in \{\pm 1\}$ the sign of an isolated solution. We can now define the degree-zero chain map

Definition 3.4.

$$\Phi_\chi^\# : CF_k(H^-, J^-) \longrightarrow CF_k(H^+, J^+), \quad \Phi_\chi^\#(\gamma^-) = \sum_{\substack{\gamma^+ \in \mathcal{P}_0(H^+) \\ \mu_{\text{CZ}}(\gamma^+) = k}} \sum_{u \in \mathcal{N}_0(\gamma^-, \gamma^+; \chi)} \epsilon^\chi(u) \gamma^+.$$

The coefficient field is $\mathbb{Q}$ here; all sums are finite by the preceding compactness statement. Orientations of generators and determinant lines are chosen compatibly with the Floer differential.

Let $\partial^\pm$ be the Floer differentials at the two ends. To see why $\partial^+ \Phi_\chi^\# = \Phi_\chi^\# \partial^-$, take $\mu_{\text{CZ}}(\gamma^-) = k$ and $\mu_{\text{CZ}}(\gamma^+) = k - 1$. The moduli space $\mathcal{N}_1(\gamma^-, \gamma^+; \chi)$ has dimension one. Its compactification is a

compact oriented one-manifold whose boundary consists of two kinds of broken trajectory:

$$\begin{aligned} \partial \overline{\mathcal{N}}_1(\gamma^-, \gamma^+; \chi) = & \bigcup_{\mu_{\text{CZ}}(y)=k} \mathcal{N}_0(\gamma^-, y; \chi) \times \widehat{\mathcal{M}}_1(y, \gamma^+; H^+, J^+) \\ & \cup \bigcup_{\mu_{\text{CZ}}(z)=k-1} \widehat{\mathcal{M}}_1(\gamma^-, z; H^-, J^-) \times \mathcal{N}_0(z, \gamma^+; \chi). \end{aligned}$$

Here $\widehat{\mathcal{M}}_1$ is the isolated index-one ordinary Floer moduli space *after* quotienting by s translation. The first kind contributes the coefficient of γ^+ in $\partial^+ \Phi_\chi^\#(\gamma^-)$, and the second contributes that in $\Phi_\chi^\# \partial^-(\gamma^-)$. The oriented boundary identity, with the coherent orientation convention, gives their difference zero. More explicitly, if y is an intermediate orbit of index k , a zero-dimensional continuation cylinder from γ^- to y can be followed by an ordinary index-one cylinder from y to γ^+ . The other possibility starts with an ordinary index-one cylinder from γ^- to an index- $(k-1)$ orbit z and finishes with a zero-dimensional continuation cylinder from z to γ^+ . Every end of the one-dimensional moduli space is of one of these forms under the compactness hypotheses. Conversely, gluing supplies an end for each regular broken pair. The boundary orientation assigns opposite algebraic contributions to the two sorts after the standard determinant-line identifications; merely listing the pairs without the converse gluing statement would not yet prove the chain-map identity. This is the same boundary-count mechanism that proves $\partial^2 = 0$, now with a continuation level between the two ends.

Next we show independence of the path. Let $\chi_0, \chi_1 \in \mathcal{P}_{\text{reg}}$ have the same endpoints, and choose a generic homotopy $h^r = \chi_r$, $r \in [0, 1]$, fixed near $r = 0, 1$ and at both s ends. Define the parameterized moduli space

Definition 3.5.

$$\mathcal{O}(\gamma^-, \gamma^+; h) = \{(r, u) : r \in [0, 1], u \in \mathcal{N}(\gamma^-, \gamma^+; \chi_r)\}.$$

For a regular parameterized section its expected dimension is

$$\dim \mathcal{O}(\gamma^-, \gamma^+; h) = \mu_{\text{CZ}}(\gamma^-) - \mu_{\text{CZ}}(\gamma^+) + 1.$$

Its compactification has parameter-end faces and, in dimension one, broken Floer ends. It is not generally a compact manifold without these added strata.

When $\mu_{\text{CZ}}(\gamma^-) = k$ and $\mu_{\text{CZ}}(\gamma^+) = k+1$, the parameterized space is zero-dimensional. Its signed count defines a degree-*plus-one* operator

$$P_h : CF_k(H^-, J^-) \longrightarrow CF_{k+1}(H^+, J^+), \quad P_h(\gamma^-) = \sum_{\substack{\mu_{\text{CZ}}(\gamma^+)=k+1 \\ (r,u) \in \mathcal{O}_0(\gamma^-, \gamma^+; h)}} \epsilon^h(r, u) \gamma^+.$$

For equal endpoint indices the parameterized moduli space has dimension one. Its boundary includes $r = 0$, $r = 1$, and breaks at the minus or plus ends. Counting these oriented boundary points yields

$$\Phi_{\chi_1}^\# - \Phi_{\chi_0}^\# = \partial^+ P_h + P_h \partial^-. \quad (8)$$

Thus the two maps induce the same map on homology. The sign between the parameter-end contributions depends on the orientation of $[0, 1]$, and (8) fixes our choice. This is the chain homotopy statement; it does not say that the two chain maps are equal or that their difference is itself a single boundary.

The degree in this construction is easy to overlook. Adding the homotopy parameter raises the expected dimension by one. To count isolated parameterized solutions, the unparameterized

continuation index must therefore be -1 , which makes the target generator have index $k + 1$. Hence P_h has degree $+1$, as required by (8), since each differential has degree -1 . When the parameterized index is 0, its one-dimensional compactification has four sources of ends: $r = 0$, $r = 1$, breaking before the continuation cylinder, and breaking afterward. The first two produce the left side of (8); the other two produce its right side. This also shows why the homotopy must remain fixed near the two parameter endpoints, so that its parameter boundary is exactly the two selected continuation maps.

Finally, consider three regular pairs (H^-, J^-) , (H^0, J^0) , and (H^+, J^+) , with a path χ_0 from $-$ to 0 and a path χ_1 from 0 to $+$. Form a long-neck concatenation $\chi_{01,R}$, placing the transition of χ_0 near $s = -R$ and that of χ_1 near $s = R$, with a constant middle region. For large R this is a smooth path after matching the constant ends; no unsupported exact equality of chain maps is claimed. The gluing theorem identifies the zero-dimensional solutions near a two-level broken configuration with solutions for a long neck. Varying R gives a chain homotopy between the concatenation map and $\Phi_{\chi_1}^\# \Phi_{\chi_0}^\#$. Therefore, on homology,

$$\Phi_{-+} = \Phi_{0+} \circ \Phi_{-0}.$$

A constant continuation datum induces the identity: its index-zero solutions are the constant cylinders, while nonconstant translation orbits cannot contribute isolated zero-dimensional points. Equivalently a short homotopy gives the same identity by the local continuation calculation. Apply composition to a path and a reversed path. Both concatenations are homotopic, with fixed endpoints, to constant paths, so

$$\Phi_{+-} \circ \Phi_{-+} = \text{id}_{HF_*(H^-, J^-)}, \quad \Phi_{-+} \circ \Phi_{+-} = \text{id}_{HF_*(H^+, J^+)}.$$

This completes the proof of Theorem 3.1 using the stated compactness and gluing theorems.

Proof of Theorem 1.1 in the aspherical case. Choose a Morse function H and a compatible J as in Theorem 2.8. For sufficiently large k , Proposition 2.1 and the low-index comparison identify the Floer complex of H/k with the shifted Morse complex. By Morse homology and Theorem 3.1, for any nondegenerate Hamiltonian K ,

$$HF_j(K; \mathbb{Q}) \cong HF_j(H/k, J; \mathbb{Q}) \cong H_{j+n}(M; \mathbb{Q}).$$

Taking total dimensions, the number of generators of $CF_*(K; \mathbb{Q})$ is $\#\mathcal{P}_0(K)$, so

$$\#\mathcal{P}_0(K) = \sum_j \dim CF_j(K; \mathbb{Q}) \geq \sum_j \dim HF_j(K; \mathbb{Q}) = \sum_{i=0}^{2n} \dim H_i(M; \mathbb{Q}).$$

Only finitely many degrees occur in the aspherical grading, and nondegeneracy ensures the set of generators is finite. $\square$

Remark 3.6 (Monotone manifolds and the full statement). For a monotone manifold, $[\omega]|_{\pi_2} = \lambda c_1(TM)|_{\pi_2}$ with $\lambda > 0$, the preceding bare chain identification and compactness arguments require modification. Periodic orbits carry cappings, recapping changes both action and Conley–Zehnder grading, and holomorphic spheres can occur in compactifications. One uses a Novikov coefficient field or ring and the appropriate monotone Floer theory, with the requisite transversality and compactness, to define continuation maps. The Piunikhin–Salamon–Schwarz (PSS) comparison identifies Hamiltonian Floer homology with quantum homology as modules over the Novikov coefficients; the underlying quantum module is ordinary homology tensored with those coefficients [7, 2]. Its rank is therefore $\sum_i \dim H_i(M; \mathbb{Q})$ when the chosen coherent orientation theory permits rational coefficients.

The Floer chain module has rank equal to the number of geometric contractible periodic orbits. This gives the same inequality for monotone manifolds. The general closed-manifold version of Theorem 1.1 uses further virtual or equivalent foundational techniques [5, 6]; our direct small-Hamiltonian proof establishes the aspherical case and the cited Floer–quantum comparison supplies the monotone case. The presence of possible sphere bubbles is the specific reason the earlier unqualified compactness assertion cannot serve as a proof in the monotone case.

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